

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

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Abstract

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account in an emotional Context Maintenance and Retrieval (eCMR) model, in which context reinstatement, competitor encoding, and retrieval control produce the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state binds task representations to overlapping context. When later retrieval contexts cue both film items and task competitors, uncontrolled retrieval from context cues is less likely to sample trauma-film items in intrusion-like retrieval, even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals and control processes reinstate film context and prioritize film items over competitors for retrieval; film-related cues can also steer retrieval toward film context across test formats. Together, the simulations reproduce selective impairment of intrusion-like retrieval and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both uncontrolled and controlled access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding or after a later film reminder, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Asselbergs et al., 2023](#); [Deepprose et al., 2012](#); [Holmes et al., 2009, 2010](#); [James et al., 2015](#); [Kessler et al., 2020](#); [Lau-Zhu et al., 2019](#)). Even for memories of real-life traumatic

events, including motor vehicle accidents, emergency caesarean section, and war-related trauma, the same post-reminder intervention can reduce intrusions while leaving deliberate recall relatively intact ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent interpretation treats the selective interference effect as evidence for separable memory representations ([Brewin et al., 1996](#); [Brewin, 2014](#)). Under this dual-representation account, traumatic experience produces distinct sensory and contextual memory traces — the first supporting involuntary re-experiencing, the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). A visuospatial interference task competes for the perceptual processing resources needed to consolidate the sensory trace, reducing later intrusions while leaving the contextual trace and the voluntary memory it supports relatively intact ([Holmes et al., 2009](#); [James et al., 2015](#)). When the intervention follows a reminder at longer delays, the account invokes reconsolidation: the reminder reactivates the sensory trace, and the interfering task disrupts its restabilization during a time-limited window ([James et al., 2015](#); [Kindt et al., 2009](#)). The account thus ties selective interference to separate memory representations, modality-specific disruption of one trace, and a discrete reconsolidation window.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation, the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal within a retrieved-context framework drawing on the emotional Context Maintenance and Retrieval model family, including eCMR and CMR3 (Cohen & Kahana, 2022; Talmi et al., 2019). Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009). Emotional extensions add a source-context stream for emotional and neutral states and an emotion-dependent binding strength, allowing the same architecture to capture how emotional content changes the organization and success of recall (Cohen & Kahana, 2022; Talmi et al., 2019). Variants also model strategic control over memory search, allowing the model to capture how retrieval goals and task demands shape recall (Cohen & Kahana, 2022; Lohnas et al., 2015; Sederberg et al., 2008). Together, these commitments make the framework a principled setting for testing whether selective interference can arise within a single emotional episodic-memory system.

For selective interference, our model treats the film, any reminder, and subsequent interference as episodes encoded by binding item representations to a continuously evolving context representation (Howard & Kahana, 2002; Polyn et al., 2009; Yonelinas et al., 2019). Post-film or post-reminder activity can therefore encode task items in context states that overlap with film context, including states reinstated by reminders. At retrieval, the current context state cues memory search, so competition depends on how strongly film and task episodes are associated with similar context states. Deliberate recall can reduce this competition when retrieval begins from a more film-directed state and when selection favors film items over competitors. The account therefore attributes the intrusion-voluntary-memory dissociation to different retrieval states applied to shared memory associations. This reframes selective interference as a change in the competitive retrieval environment around the film, rather than as a change in the availability of the film representation after encoding or reminder reactivation.

Our simulations evaluate the account at the level where a retrieved-context model makes its distinctive predictions: the distribution and organization of recall across the film, interference, and delay phases of a trauma-film paradigm. They first demonstrate the core selective-interference

dissociation without separately vulnerable representations: an interference task can reduce intrusion-like retrieval more than deliberate recall, even though the same stored film episode supports both. They then distinguish the control mechanisms that protect deliberate recall, showing how a more film-directed retrieval state and a sharper decision rule produce different patterns of impairment in both recall rates and recall organization. The next simulations separate the contributions of competitor strength and contextual overlap, clarifying when an interference task becomes an effective competitor for film retrieval and how reminder-driven reinstatement can support delayed interference. Follow-up simulations use the same framework to interpret heterogeneity across empirical paradigms. They show that film-related cues during intrusion monitoring can narrow the intrusion-voluntary-memory dissociation by making intrusion-like retrieval less purely unguided. They also show that tests supplying a film item can be less vulnerable to interference, because the supplied item can reinstate its associated context instead of requiring retrieval to sample among film items and task competitors from a broad contextual cue. The result is a mechanistic alternative to separately vulnerable memory representations and a set of specific hypotheses about when selective interference should appear and how it should vary across paradigms.

A Retrieved-Context Account of Selective Interference

The simulations rely on three retrieved-context mechanisms. Context-to-item retrieval uses current context to cue candidate items. Competition follows when film items and task competitors are associated with similar context states. Item-to-context retrieval allows recalled items, reminder cues, and supplied film items to reinstate associated context. Film-directed retrieval control changes the starting cue and selection rule used to favor film items over task competitors. Figure 1 shows how these mechanisms map onto the trauma-film paradigm.

Contextual similarity is established during encoding. Film items and task items are distinct item representations, but each is linked to the context state active when it is encountered. Task items are most likely to share film-relevant context when they are encoded soon after the film, after a reminder has reinstated film-associated context, or under matching source/emotional

Figure 1

A retrieved-context account of selective interference. The model explains intrusion-voluntary-memory dissociations by applying a single retrieved-context architecture to the phase structure of post-encoding and post-reminder interference paradigms. (A) The simulated paradigm includes film encoding, optional reminder-driven context reinstatement, interference-task encoding, filler activity, and later retrieval. (B) Encoding links film and task items to the context states active during their phases; task-associated context is most similar to film-associated context when task encoding occurs soon after the film, after reminder-driven reinstatement, or under matching source/emotional context. (C) Feature-to-context memory (M^{FC}) and context-to-feature memory (M^{CF}) implement the retrieved-context cycle, allowing items and reminders to reinstate associated context and allowing current context to cue competing candidate items. (D) Uncontrolled retrieval from ongoing context and deliberate recall operate over the same stored associations, but deliberate recall can be protected by film-directed cue construction and competition focusing.

conditions. Delay and filler activity move context away from the film, reducing the likelihood that later retrieval cues film items and task competitors together. These conditions determine whether the task adds effective competitors to the film's retrieval environment.

Context-to-item retrieval uses the current context state to cue candidate items. When a retrieval context resembles states associated with both film items and task items, the same cue supports both sets. Uncontrolled retrieval from ongoing context is vulnerable to this competition: as task competitors receive support, any given film item becomes less likely to be sampled as an intrusion. The size of this effect depends on the number of competitors, their learned support, and the similarity between their associated temporal or source/emotional context states and film context. Perceptual modality contributes through those quantities: a visuospatial task is effective to the extent that it creates competitors with sufficient learned support or sufficiently similar associated context.

Item-to-context associations allow recalled items, reminder cues, and supplied film items to reinstate associated context. In ordinary recall, a retrieved item shifts context toward the state in which that item was encoded, making nearby studied items more likely to be retrieved next. This context-updating loop is the standard retrieved-context explanation of temporal contiguity and temporally organized recall across filled delays ([Howard & Kahana, 2002](#); [Kahana, 1996](#); [Sederberg et al., 2008](#)). The same dynamics allow recent but irrelevant experiences to intrude during recall of a target episode, with intrusion frequency decaying gradually over intervening time ([Howard et al., 2008](#); [Lohnas et al., 2015](#); [Zaromb et al., 2006](#)). In delayed-interference paradigms, a reminder of the film uses this same pathway to reinstate film-associated context before task encoding ([James et al., 2015](#); [Kessler et al., 2020](#)). Delayed interference is therefore graded by reminder strength and subsequent drift, because those quantities determine how much film-associated context remains available when the task is encoded. Film-related cues during monitoring and tests that supply a film item use the same item-to-context pathway during later access. They can pull retrieval context toward film-associated states or retrieve a supplied item's associated context, reducing reliance on broad context-to-item sampling. Interference items encoded later can add competitors to context-to-item retrieval, but they do not directly overwrite a film item's own item-to-context association. Recognition-like access is therefore less exposed to the contextual-competition mechanism that reduces uncontrolled retrieval from ongoing context, while deliberate free recall remains a controlled version of context-to-item search.

Deliberate recall is protected only to the extent that retrieval control changes the search. Like uncontrolled retrieval, deliberate recall uses context-to-item retrieval and remains exposed to competitor interference. The first control lever is cue construction or start-state control, which reinstates film-associated context before recall and biases search toward film items. The second is competition focusing or filtering, which sharpens selection among candidates and favors strongly supported film items over weaker or off-target competitors. These levers yield graded protection: deliberate recall can lose film items when competition is strong, but control can reduce that loss relative to uncontrolled retrieval from ongoing context.

Emotional extensions of CMR add a source-context stream that can distinguish items encoded in emotional versus neutral states (Cohen & Kahana, 2022; Talmi et al., 2019). In the present source-only variant, this stream represents whether film and task items are associated with matching or different source/emotional context states. The formal model specified next instantiates these commitments as phase-specific encoding, reminder-driven reinstatement, retrieval-mode and control settings, and source/emotional overlap.

Model and Paradigm

Representations and Memory Architecture

The model represents each study event as an orthonormal item vector f_i . The selective-interference paradigm presents items across a fixed phase sequence: film, break, reminder, interference, filler, and retrieval. Film items are the target items whose later access indexes intrusion-like or voluntary memory. Interference items are later task events that can become competitors for film retrieval. Each item also carries a source flag $e_i \in \{0, 1\}$ indicating whether it belongs to the source/emotional class manipulated in a given simulation, and a target flag $t_i \in \{0, 1\}$ indicating whether it is a film item.

Two context streams evolve over the experiment. Temporal context c^T is a unit-length vector with a start-of-list unit and one unit for each item. Source context c^S is a unit-length vector with start, emotional/source, and neutral/source dimensions. Both context streams begin in their start states. As items are encoded or retrieved, context drifts toward the context associated with the current item.

Four associative matrices link items to contexts. $M^{FC,T}$ maps item features to temporal context and supports item-to-context reinstatement. $M^{CF,T}$ maps temporal context to item features and supports context-to-item competition. $M^{FC,S}$ and $M^{CF,S}$ provide the corresponding item-to-source-context and source-context-to-item mappings. The source pathway is the source-only eCMR extension used here: it can add source-based support for retrieval without making emotional/source status a hidden modulation of temporal learning.

Table 1 lists the model structures and parameters used throughout the simulations.

Pre-Experimental Associations

Pre-experimental initialization establishes baseline associations before the experiment.

For the temporal pathway:

$$M_{pre}^{FC,T}(i, j) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$M_{pre}^{CF,T}(j, i) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases}$$

The start context has no pre-experimental context-to-item support. For context units corresponding to items, δ gives each item self-support and α gives all other items shared support. This support is explicit so that simulations can test whether selective-interference effects depend on learned contextual competition rather than hidden background support.

For the source pathway, define each item's source vector as:

$$u_i = \begin{bmatrix} 0 \\ e_i \\ 1 - e_i \end{bmatrix}$$

$$M_{pre}^{FC,S}(i, k) = (1 - \gamma)u_{ik}$$

$$M_{pre}^{CF,S}(k, i) = 0$$

The source pathway therefore begins with item-to-source associations but no source-to-item competition. Source-based competition is learned during encoding and is controlled by the source learning scale.

Encoding Across Phases

The film, break, interference, and filler phases use the same encoding operation with phase-specific rates. Let p index the current phase and let β_p and λ_p denote that phase's temporal drift rate and context-to-item learning scale. The film phase encodes target items with baseline rates. The break and filler phases add intervening experience. The interference phase encodes task items whose effect depends on their strength and the similarity between their associated context states and film context.

For item i , temporal context input is retrieved by probing item-to-context memory:

$$c_i^{IN,T} = M^{FC,T} f_i$$

The model normalizes the input to unit length. Temporal context then drifts toward the normalized input:

$$c_i^T = \rho_i^T c_{i-1}^T + \beta_p c_i^{IN,T}$$

$$\rho_i^T = \sqrt{1 + \beta_p^2 \left[\left(c_{i-1}^T \cdot c_i^{IN,T} \right)^2 - 1 \right]} - \beta_p \left(c_{i-1}^T \cdot c_i^{IN,T} \right)$$

The ρ_i^T term preserves unit length. The drift rate β_p controls how far the current phase moves context away from the prior phase. In selective-interference simulations, this rate helps determine whether interference items are encoded near or far from the film context.

Feature-to-context associations update by an outer product after context updating:

$$\Delta M^{FC,T} = \gamma f_i (c_i^T)^\top$$

A primacy function weights temporal context-to-item learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1$$

$$\Delta M^{CF,T} = \lambda_p \phi_i c_i^T f_i^\top$$

Here, λ_p is the phase-specific MCF learning scale. Increasing λ_p during interference makes task items stronger competitors for later context-to-item retrieval. The primary source-only eCMR variant does not allow source/emotional status to modulate temporal context-to-item learning; temporal learning remains primacy-driven unless a later sensitivity analysis explicitly turns on broad temporal modulation.

Source/Emotional Context

Source context updates in parallel with temporal context. For item i , source-context input is retrieved by probing $M^{FC,S}$:

$$c_i^{IN,S} = M^{FC,S} f_i$$

Source context then integrates this input:

$$c_i^S = \rho_i^S c_{i-1}^S + \beta_{enc}^S c_i^{IN,S}$$

Source feature-to-context learning mirrors the temporal pathway:

$$\Delta M^{FC,S} = \gamma f_i (c_i^S)^\top$$

Source context-to-item learning is gated by the source flag:

$$\Delta M^{CF,S} = \lambda_p \phi_i \omega_S e_i c_i^S f_i^\top$$

The parameter ω_S controls source-context-to-item learning for source/emotional items. When $\omega_S = 0$ or when all items are neutral, the source pathway does not contribute source-to-item competition. When film and interference items share source/emotional status, the source pathway can increase overlap between the context cues supporting target and competitor items beyond temporal proximity alone.

Each phase applies multiplicative scale factors to the base encoding rates. The film phase uses base rates directly. Subsequent phases modulate drift rate, which controls context proximity to preceding items, and M^{CF} learning scale, which controls competitor encoding strength.

The interference scales s_{int}^{β} and s_{int}^{ϕ} are central to the simulations. The drift scale controls how much temporal context changes during interference encoding. The MCF scale controls competitor encoding strength: stronger competitors capture more retrieval probability from film targets. Together, drift and learning scales define whether interference is merely added experience or a set of strong competitors encoded into film-related context.

Reminder Reinstatement

In the trauma-film paradigm, a film clip presented as a reminder between the break and interference phases serves to reinstate the film's encoding context. The model implements this in two steps. First, both context representations drift toward their respective start-of-list states:

$$c_{\text{rem}}^T = \rho c^T + \beta_{\text{rem,start}} c_0^T$$

$$c_{\text{rem}}^S = \rho_S c^S + \beta_{\text{rem,start}} c_0^S$$

Here, $\beta_{\text{rem,start}} = s_{\text{rem,start}}^{\beta} \cdot \beta_{\text{start}}$ sets how strongly context shifts toward the pre-experimental state before the reminder arrives. This partial reinstatement moves context back toward temporal states associated with film encoding.

Second, each reminder item reinstates its specific temporal and source context through the same item-to-context pathway used during recall:

$$c_{\text{cue}}^{IN,T} = M^{FC,T} f_{\text{cue}}$$

$$c^T \leftarrow \rho_T c^T + \beta_{\text{rem}} c_{\text{cue}}^{IN,T}$$

$$c_{\text{cue}}^{IN,S} = M^{FC,S} f_{\text{cue}}$$

$$c^S \leftarrow \rho_S c^S + \beta_{\text{rem}} c_{\text{cue}}^{IN,S}$$

Here, $\beta_{\text{rem}} = s_{\text{rem}}^\beta \cdot \beta_{\text{enc}}$ controls how strongly the reminder pulls context toward the film's encoding state. Critically, reminders reinstate context without forming new associations: $M^{FC,T}$, $M^{CF,T}$, $M^{FC,S}$, and $M^{CF,S}$ remain unchanged. This distinction mirrors the clinical logic of a reminder cue — it reactivates the memory without adding new content.

The reminder mechanism has a direct consequence for interference. By pulling context back toward film-associated states, the reminder makes interference items encoded immediately afterward associated with temporal and potentially source context similar to the film context. Stronger reinstatement (s_{rem}^β closer to 1) produces more overlap between the context cues supporting film targets and interference competitors. Delayed-interference simulations therefore manipulate reminder reinstatement as a graded overlap mechanism rather than as an all-or-none reactivation window.

Retrieval and Termination

Context-to-item retrieval underlies both intrusion-like retrieval and deliberate free recall. At the start of retrieval, both context streams drift toward their initial states:

$$c_{\text{start}}^T = \rho c_N^T + \beta_{\text{start}} c_0^T$$

$$c_{\text{start}}^S = \rho_S c_N^S + \beta_{\text{start}} c_0^S$$

Here, β_{start} controls how much start-of-list context is reinstated at retrieval onset. The current temporal and source contexts then cue all recallable items through $M^{CF,T}$ and $M^{CF,S}$:

$$a_i^T = [M^{CF,T} c^T]_i r_i$$

$$a_i^S = [M^{CF,S} c^S]_i r_i$$

$$a_i = a_i^T + a_i^S$$

Here, r_i is a binary mask indicating whether item i is currently recallable. Because all recallable items contribute to the denominator, encoding additional competitors in contextually similar states dilutes each film item's share of retrieval probability. The choice sensitivity τ power-scales the combined activation vector, with a small lower bound ϵ applied for numerical stability:

$$A_i = (a_i^\tau + \epsilon) r_i$$

The probability of recalling item i then follows a Luce choice rule:

$$P(i \mid \text{recall}) = \frac{A_i}{\sum_k A_k}$$

At each retrieval step, the model may terminate with probability p_{stop} or recall an item with probability $(1 - p_{\text{stop}}) \cdot P(i \mid \text{recall})$. Termination probability grows with recall position:

$$p_{\text{stop}}(j) = \theta_s e^{\theta_r \cdot j}$$

Here, j is the number of items already recalled, θ_s sets the baseline termination probability, and θ_r controls its growth. When an item is recalled, both context streams reinstate its associated context:

$$c^T \leftarrow \rho_T c^T + \kappa_i \beta_{\text{rec}} M^{FC,T} f_i$$

$$c^S \leftarrow \rho_S c^S + \kappa_i \beta_{\text{rec}}^S M^{FC,S} f_i$$

The target-monitoring scale κ_i equals 1 for film items and can be reduced for non-film competitors. This target monitoring setting represents a retrieval policy in which rejected competitors have less influence on the next context state. This reinstatement drives context toward the recalled item's encoding neighborhood, biasing subsequent retrievals toward temporally and emotionally proximate items. This mechanism produces the temporal organization expected of retrieved-context models.

Retrieval Modes and Control Settings

The simulations vary retrieval mode and film-directed control settings without changing the stored film and task episodes. Intrusion-like retrieval is modeled as relatively broad context-to-item sampling from the current retrieval context. Deliberate recall uses the same context-to-item pathway, but with stronger film-directed start reinstatement, sharper selection among candidates, and optionally reduced context updating after rejected non-film competitors. These settings do not create a separate voluntary-memory store; they change the retrieval state and selection policy applied to the same item-context associations.

Probe-driven access uses the item-to-context pathway to reduce reliance on broad context-to-item search. When a test supplies a film item, that item can retrieve its own associated context through item-to-context memory:

$$c_i^{IN,T} = M^{FC,T} f_i$$

This route supplies more target-specific information than broad context-to-item search. It should therefore be less exposed to competitors encoded near the film context, although the manuscript treats recognition-like access as a probe-driven case to be interpreted alongside the task design rather than as an absolute recognition result.

Parameters and Simulation Factors

The base parameter vector is anchored by external eCMR fits rather than fit to the selective-interference simulations themselves. For these simulations, the anchor is a pooled source-only eCMR fit to the Ukraine war photographs free-recall data ([Dupertuys et al., 2026](#)).

The selective-interference simulations then manipulate paradigm-level scale factors that determine phase drift, competitor learning, source/emotional overlap, reminder reinstatement, and retrieval control. This separation keeps fitted memory-search parameters distinct from the theoretical manipulations used to create selective interference.

Table 3 lists the main scale factors and their theoretical roles.

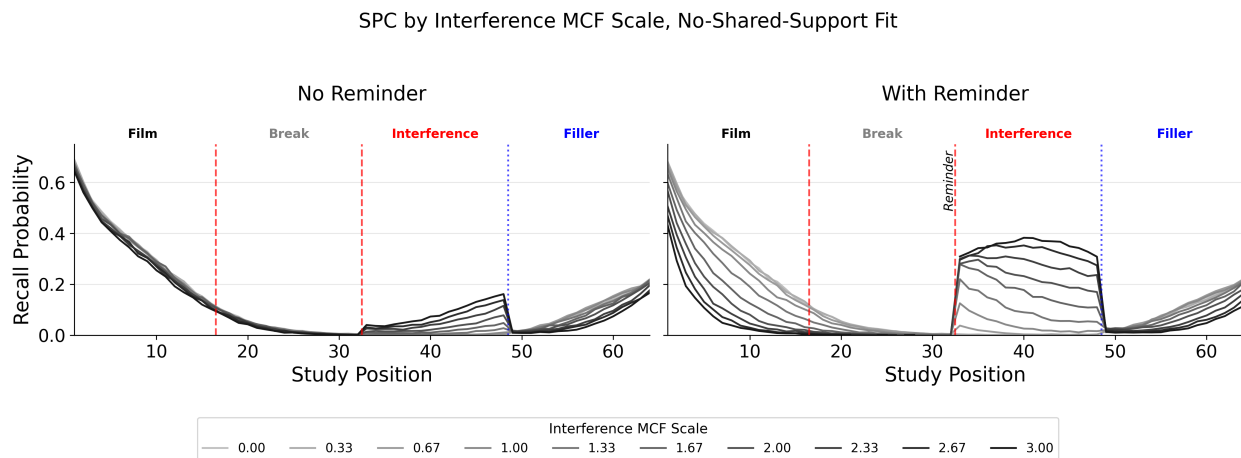
The paradigm geometry mirrors trauma-film selective-interference designs at an abstract level. Each simulated trial includes film, optional break, reminder, interference, and filler phases, with phase lengths and phase-specific scaling set by the simulation factors in Table 3. The main simulation outputs track which phase each recalled item came from, how often film items are recalled, and how those quantities change across interference and retrieval-control manipulations. Because retrieval proceeds as a sequence of competitive samples terminated by a growing stop probability, the expected number of film items recalled depends on how much retrieval probability competitors capture at each step. These dependent measures provide a bridge between the model's formal predictions and the empirical phenomena the simulations target.

Simulations

The first rendered sweep included here orients the reader to the simulated phase structure before the later sections develop the final manuscript contrasts. These plots are diagnostic outputs from the interference-strength sweep, not stand-alone evidence for the full intrusion-voluntary-memory dissociation.

Figure 2

Phase-resolved recall in the interference-strength sweep. Serial-position curves show recall probability across the film, break, interference, and filler phases of the simulated paradigm, separately for no-reminder and reminder conditions. Darker lines indicate stronger context-to-item learning for interference-task items. The figure shows how the interference manipulation changes the distribution of retrieval across encoded phases, rather than only changing total film recall.



Simulation 1: Core Selective-Interference Dissociation

Premise: The basic selective-interference result is a 2 x 2 contrast between interference condition and retrieval mode. The section asks whether the same stored film episode and the same post-film interference manipulation can produce larger impairment in intrusion-like retrieval than in deliberate recall.

Intended text take-away: This section establishes the core sufficiency result. The model can reproduce the qualitative intrusion-voluntary-memory dissociation without separately weakening film items for one measure. Deliberate recall may still be impaired; the claim is relative protection, not immunity.

Simulation 2: Control Mechanisms Underlying Deliberate-Recall Protection

Premise: Deliberate recall is not categorically protected. Its relative sparing depends on control operations that make retrieval more film-directed.

Intended text take-away: This section earns the “relatively intact voluntary memory” claim. Sparing is a graded consequence of retrieval control rather than a property attached to voluntary tests as a category. Weakening these control operations should make deliberate recall resemble intrusion-like retrieval more closely.

Simulation 3: Competitor Strength, Contextual Overlap, and Emotional-Context Amplification

Premise: An interference task becomes effective when it encodes competitors with enough learned support and with associated context states similar enough to film context. This section asks what makes a task Tetris-like in model terms.

Intended text take-away: This section explains what makes interference effective. The account should not reduce to generic distraction, list length, or task difficulty. Competitor strength matters because competitors are retrievable from film-related context, and shared source/emotional context is relevant only insofar as it increases that competition.

Simulation 4: Reminder Reinstatement and Delayed Interference

Premise: The delayed-reminder case applies the overlap mechanism after context has drifted away from the film. A reminder matters because it reinstates film context before competitor encoding.

Intended text take-away: This section connects the model to post-reminder interventions. The reminder is not treated as a special storage-disruption device. It restores the contextual condition required for later competitor encoding to affect film-item retrieval.

Simulation 5: Heterogeneity Across Recall-Like Test Formats

Premise: Intrusion diaries, vigilance-style monitoring, free recall, and cued recall differ in how much they rely on broad context-to-item search versus film-directed retrieval support. This section maps the model’s retrieval-control mechanisms onto variation across recall-like empirical measures.

Intended text take-away: This section interprets heterogeneity across empirical paradigms without treating intrusion and voluntary memory as fixed computational kinds. The model

predicts a continuum of vulnerability across recall-like tests depending on how much the task supports film-directed retrieval.

Simulation 6: Probe-Driven Access in Recognition-Like Tests

Premise: Tests that supply a film item are a limiting case of voluntary access because they begin closer to the target item than broad recall does. This section preserves the useful part of the old recognition result without treating probe-driven access as categorically protected.

Intended text take-away: This section handles tests that supply a film item and recognition-like measures as a probe-driven access case. The claim is that probe-driven access reduces exposure to contextual competition, not that recognition escapes the mechanism in every design or that all voluntary memory has the same protection.

General Discussion

References

- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>
- Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686. <https://doi.org/10.1037/0033-295X.103.4.670>
- Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117(1), 210–232. <https://doi.org/10.1037/a0018113>
- Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders. *Psychological Review*, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>

- Deeprise, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2), 758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>
- Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.
- Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)
- Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566. <https://doi.org/10.1016/j.actpsy.2007.11.002>
- Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>
- Holmes, E. A., James, E. L., Coode-Bate, T., & Deeprise, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>
- Holmes, E. A., James, E. L., Kilford, E. J., & Deeprise, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>
- Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research and Therapy*, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299. <https://doi.org/10.1006/jmps.2001.1388>
- Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory:

- Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.
- Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R., Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682.
<https://doi.org/10.1038/mp.2017.23>
- Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L. (2019). Intrusive memories of trauma: A target for research bridging cognitive science and its clinical application. *Clinical Psychology Review*, 69, 67–82.
<https://doi.org/10.1016/j.cpr.2018.08.005>
- James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., & Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215.
<https://doi.org/10.1177/0956797615583071>
- James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016). The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142.
<https://doi.org/10.1016/j.cpr.2016.04.010>
- Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1), 103–109.
- Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E. A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>
- Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K., Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020). Visuospatial computer game play after memory reminder delivered three days after a

traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454.

<https://doi.org/10.1016/j.jbtep.2019.01.006>

Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258.

<https://doi.org/10.1038/nn.2271>

Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.

<https://doi.org/10.1002/acp.1611>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>

Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363.

<https://doi.org/10.1037/a0039036>

Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156.

<https://doi.org/10.1037/a0014420>

Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.

<https://doi.org/10.1037/a0013396>

Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.

<https://doi.org/10.1037/rev0000132>

Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human*

Behaviour, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>

Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra: Psychology*, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>

Yonelinas, A. P., Ranganath, C., Ekstrom, A. D., & Wiltgen, B. J. (2019). A contextual binding theory of episodic memory: Systems consolidation reconsidered. *Nature Reviews Neuroscience*, 20(6), 364–375. <https://doi.org/10.1038/s41583-019-0150-4>

Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana, M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(4), 792–804. <https://doi.org/10.1037/0278-7393.32.4.792>

Table 1

Model structures and parameters. Base parameters are anchored by pooled source-only eCMR fits to the Ukraine war photographs free-recall data (Dupertuys et al., 2026).

Symbol	Name	Description
c^T	temporal context	Recency-weighted context stream with start and item units
c^S	source context	Source/emotional context stream with start, emotional/source, and neutral/source units
f_i	item features	Orthonormal representation of item i
e_i	source flag	1 for source/emotional items, 0 for neutral/source items
t_i	target flag	1 for film items, 0 for non-film items
$M^{FC,T}, M^{CF,T}$	temporal memories	Item $\leftrightarrow$ temporal-context associations
$M^{FC,S}, M^{CF,S}$	source memories	Item $\leftrightarrow$ source-context associations
β_{enc}	encoding drift rate	Temporal context drift during study
β_{start}	start drift rate	Start-of-list reinstatement at recall onset
β_{rec}	recall drift rate	Context drift during recall
β_{enc}^S	source encoding drift rate	Source-context drift during encoding
β_{rec}^S	source recall drift rate	Source-context drift during recall

Table 2

Phase-specific encoding rates. Drift scales multiply β_{enc} and are clipped to $[0, 1]$.

Phase	Drift rate	M^{CF} learning scale	Paradigm role
Film	β_{enc}	1	Target encoding
Break	$s_{break}^{\beta} \cdot \beta_{enc}$	s_{break}^{ϕ}	Retention interval
Interference	$s_{int}^{\beta} \cdot \beta_{enc}$	s_{int}^{ϕ}	Competitor encoding
Filler	$s_{fill}^{\beta} \cdot \beta_{enc}$	s_{fill}^{ϕ}	Pre-recall buffer

Table 3

Scale factors and their simulation connections. Drift scales are clipped to [0, 1] after multiplication.

Scale factor	Base parameter or target	Role	Simulation connection
s_{break}^{β}	β_{enc}	Break-phase drift	Delay and context separation
s_{break}^{ϕ}	ϕ_i	Break-phase competitor learning	Break-task control
$s_{\text{rem,start}}^{\beta}$	β_{start}	Start-state shift before reminder	Reminder reinstatement
s_{rem}^{β}	β_{enc}	Film-item reminder drift	Reminder reinstatement
s_{int}^{β}	β_{enc}	Interference-phase drift	Target-competitor cue overlap
s_{int}^{ϕ}	ϕ_i	Interference-phase MCF learning	Competitor strength
s_{fill}^{β}	β_{enc}	Filler-phase drift	Post-interference buffer
s_{fill}^{ϕ}	ϕ_i	Filler-phase MCF learning	Filler-task control
s_{start}^{β}	β_{start}	Retrieval start reinstatement	Film-directed recall control
s^{τ}	τ	Retrieval selection sharpness	Recall control
ω_S	source MCF learning	Source/emotional learning strength	Source/emotional cue overlap
$s_{\text{enc}}^{\beta,S}$	β_{enc}^S	Source-context encoding drift	Source/emotional cue overlap
$s_{\text{rec}}^{\beta,S}$	β_{rec}^S	Source-context recall	Source-context